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Started on	Monday, 16 September 2024, 2:26 PM
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State	Finished
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Completed on	Monday, 16 September 2024, 3:38 PM
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Time taken	1 hour 11 mins
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Marks	37.78/43.00
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Grade	87.86 out of 100.00
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Question 1

Partially correct

Mark 5.78 out of 8.00

Given the scenario below answer the following questions

Consider a computer system with a physical [memory](#) consisting of four frames (Frame 0, Frame 1, Frame 2, Frame 3) and a table to keep track of page locations. A process has a virtual address space with 8 pages (Page 0 to Page 7). The time to access [memory](#) is 1000 nanoseconds and time required to access the disk is 150 000 nanoseconds.

Given the following sequence of [memory](#) references:

1. Read Page 0
2. Read Page 1
3. Read Page 2
4. Read Page 3
5. Read Page 1
6. Read Page 4
7. Read Page 2
8. Read Page 3
9. Read Page 0
10. Read Page 4
11. Read Page 5
12. Read Page 6
13. Read Page 7
14. Read Page 5

Using the LRU page replacement policy, determine the number of page faults that occur during the execution of the program. Assume that the physical [memory](#) is initially empty.

Through the execution of the program in the diagram, complete the table below

Page Accesses	Page Fault Yes/Updated No	Which Page Frame is (Just answer with the number i.e. 0)	
0	Yes	0	✓
1	Yes	1	✓
2	Yes	2	✓
3	Yes	3	✓

1	No	-1	✗
	✓		
4	Yes	0	✓
	✓		
2	No	-1	✗
	✓		
3	No	-1	✗
	✓		
0	Yes	1	✓
	✓		
4	No	-1	✗
	✓		
5	Yes	2	✓
	✓		
6	Yes	3	✓
	✓		
7	Yes	0	✗
	✓		
5	No	-1	✗
	✓		

What is the total number of hits incurred by processing this program? 5 ✓

What is the total number of misses incurred by processing this program? 9 ✓

Note you must round off your final answers for the following 2 questions to two decimal places

Determine PHit = 0.36 ✓

Determine PMiss = 0.64 ✓

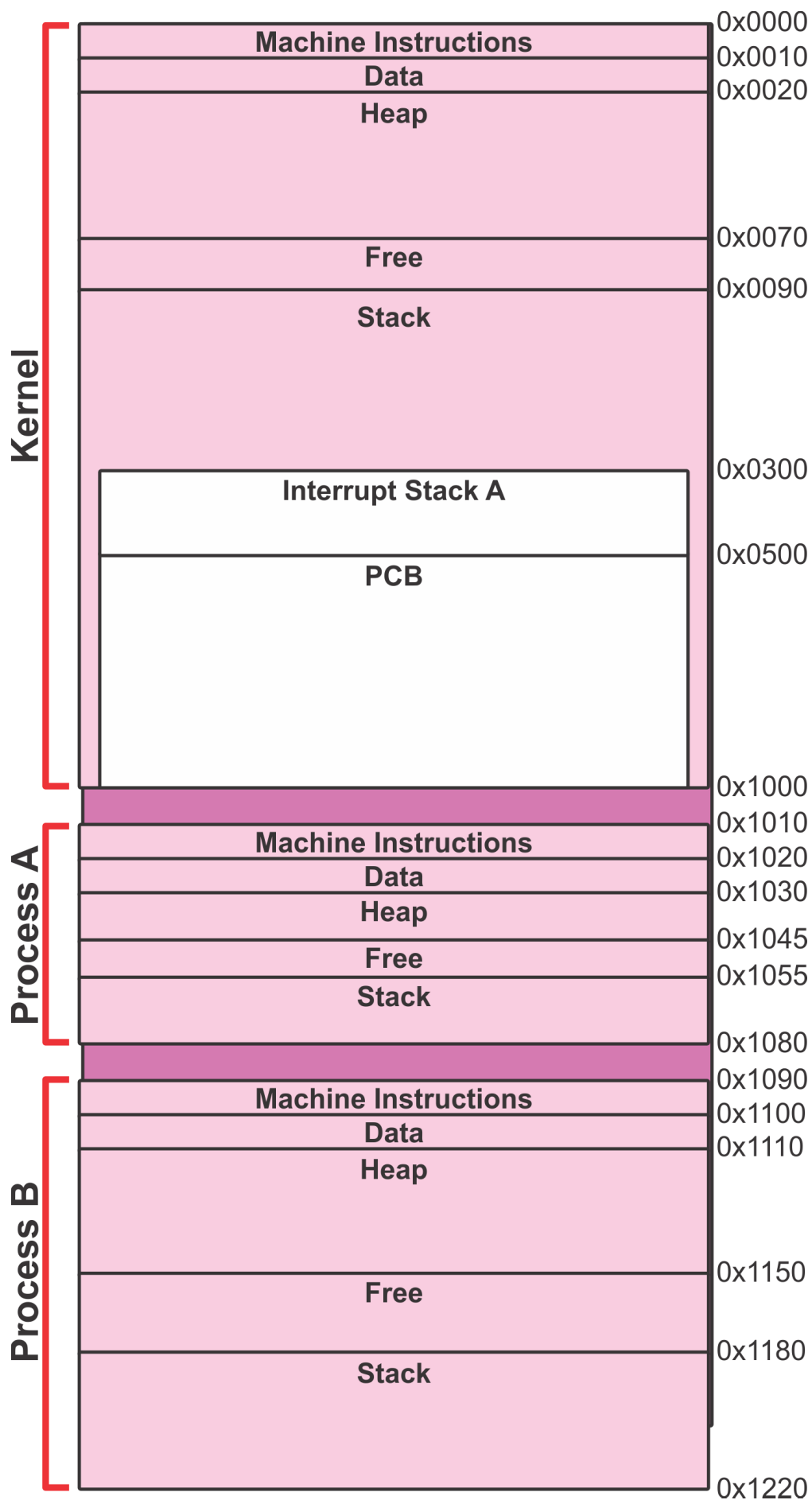
What is the AMAT for the question above leave your answer in nanoseconds and do not round off [

Question **2**

Correct

Mark 2.00 out of 2.00

Answer the following questions based on the image below, regarding the [memory](#) locations where
in the program counter and stack pointer registers during a UNIX signal initiated by Process B.



For the following questions just put the numerical value i.e. if the address is 0x1220 just put 1220. / starting and ending addresses not the base and bounds of the required region.

Which region of [memory](#) would we find the address contained in the Program Counter?

Start Address =



End Address=



Which region of [memory](#) would we find the address contained in the Stack Pointer?

Start Address =



End Address=



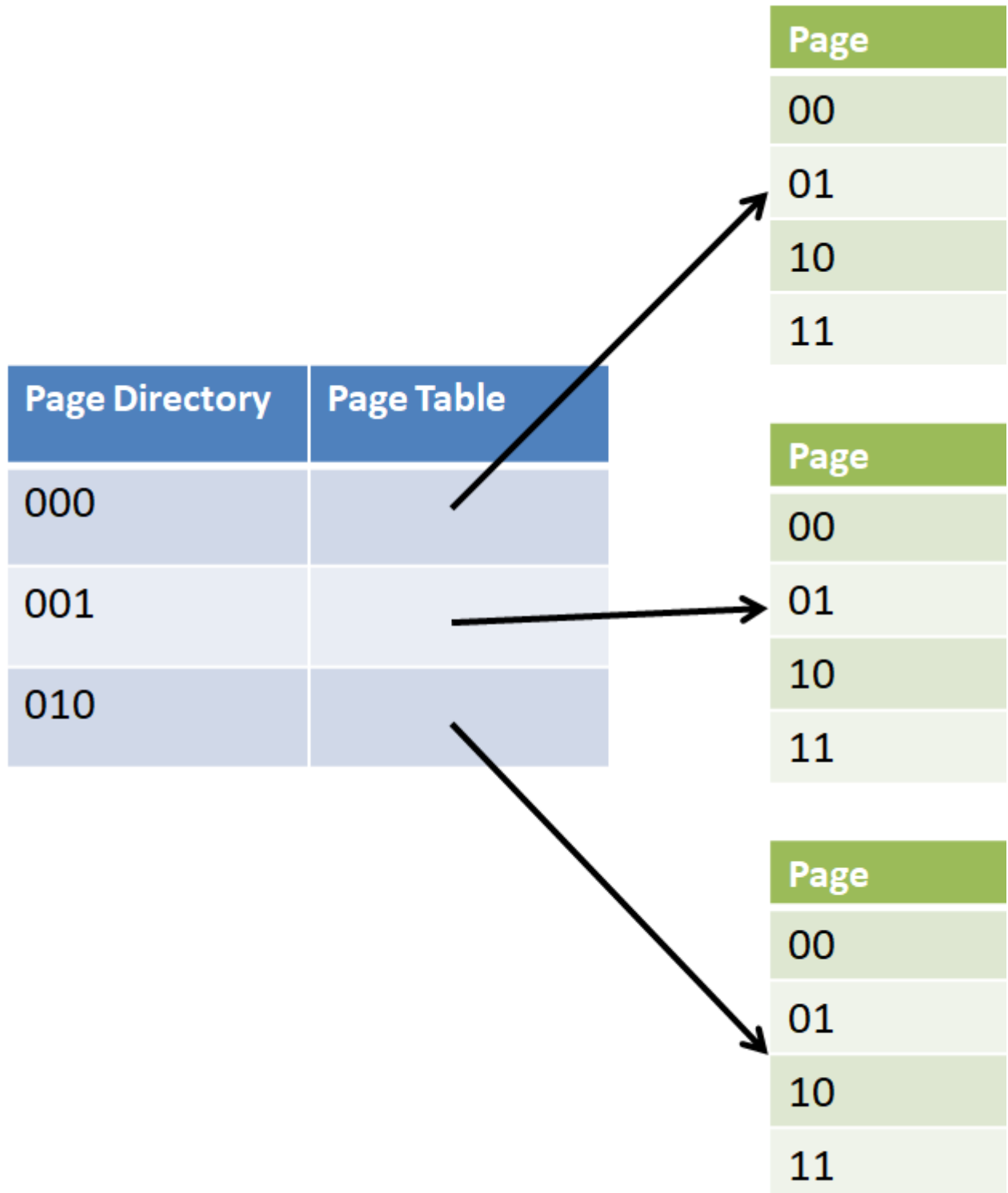
Question 3

Correct

Mark 8.00 out of 8.00

Answer the following questions related to the multilevel [paging](#) diagram depicted below in order to address: **010111111111**

- Address space is 4KB long and each page is 128bytes
- The Main Page Table contains references to 8 sub Page Tables



How many bits are required to store addresses in the virtual address space ✓

How many bits for the Offset ✓

How many bits for VPN ✓

What is the PAGE Directory in binary ✓

What is the VPN in binary ✓

How many bits are required to store the physical address ✓

What is the Physical address offset in binary ✓

What is the PFN in binary ✓

What is the entire physical address in binary ✓

Question **4**

Correct

Mark 1.00 out of 1.00

What type of Mode Transfer occurs when a USB is inserted?

Select one:

- ☐ a. Timer Interrupt
- ☐ b. System Call
- ☒ c. I/O Request ✓
- ☐ d. Processor Exception
- ☐ e. User Level Upcall

The correct answer is: I/O Request

Question 5

Correct

Mark 1.00 out of 1.00

If the parent process finishes execution before the child process, and the parent does not wait for the child process?

Select one:

- ☒ a. The child process becomes a zombie until the parent process collects its exit status. ✓
- ☐ b. The child process will never terminate
- ☐ c. The child process is immediately terminated.
- ☐ d. The child process continues running, and its exit status is automatically handled by the system.
- ☐ e. The child process takes over the role of the parent process.

The correct answer is:

The child process becomes a zombie until the parent process collects its exit status.

Question 6

Correct

Mark 1.00 out of 1.00

If we are told we have an Address space of 49MB and the virtual address has an offset made up of 16 bits. How many pages do we have per process?

Answer: ✓

The correct answer is: 12845056

Question 7

Partially correct

Mark 2.00 out of 4.00

For the following questions, you need to provide the correct order of events. You will be given a set and you will need to provide the string which corresponds with the order of events. Below is an example

1. April Fools Day
2. New Years Day
3. Christmas Day
4. Halloween

The answer you would need to submit in the block for this example would be: **2143**

Scenario 1 - Processing Cycle

1. The [memory](#) address is loaded into the address bus.
2. Data is transferred from [memory](#) to the register.
3. The data is stored in [memory](#).
4. The CPU fetches the instruction from [memory](#).

**Scenario 2 - System Call**

1. The data is returned to the process.
2. The kernel retrieves the necessary data from [memory](#).
3. The CPU registers the system call with the kernel.
4. The system call enters kernel mode.

**Scenario 3 - When an Interrupt is triggered**

1. The CPU saves the current process state.
2. The CPU jumps to the interrupt handler routine.
3. The kernel processes the interrupt.
4. The original process is restored and resumed.

**Scenario 4 - Page Fault**

1. The MMU identifies a page fault.
2. The CPU requests data from [memory](#).
3. The kernel loads the missing page from secondary storage into RAM.
4. The CPU retries the instruction and accesses the data.

2134

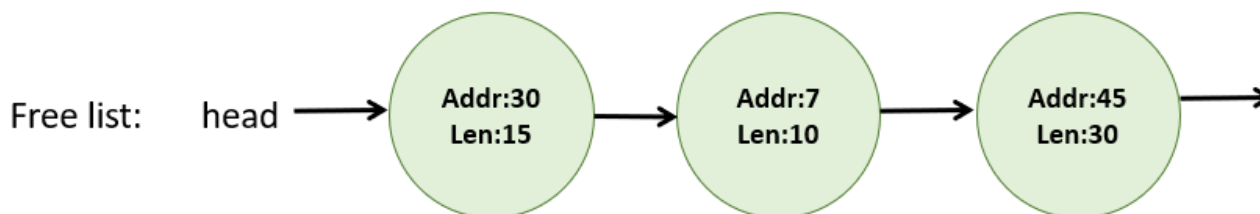


Question 8

Correct

Mark 3.00 out of 3.00

Given the diagram of the free list below answer the following questions



Scenario

- Using a [memory](#) management scheme where we always choose the largest free chunk first.
- Assuming [memory](#) is allocated to the Left-hand side of free chunks:
- When new nodes are added to the freelist they are **added to the front**.
- Allocation request sizes don't already include the header size
- The total size of each header for both free blocks and allocated blocks is of size 2
- Assuming our [memory](#) only contains addresses ranging from 0-99 (100 addresses)

Using the image above, how many addresses have been allocated already?

size = ✓

We then receive the following allocation requests in order from a user: 23, 7, 3

Answer the questions below regarding the state of the free list after all these events occur

What is the address and length of the 1st node in the free list?

Address = ✓ length = ✓

What is the address and length of the 3rd node in the free list?

Address = ✓ length = ✓

Question **9**

Correct

Mark 1.00 out of 1.00

A Timer interrupt that causes the process to be switched out is an example of an involuntary context switch.

Select one:

☒ True ✓

☐ False

The correct answer is 'True'.

Question **10**

Correct

Mark 2.00 out of 2.00

Using the buddy [memory](#) allocation system, how many times would we have to divide our [memory](#) to allocate a chunk of 120KB

Answer: 20



The correct answer is: 20

Question **11**

Partially correct

Mark 3.00 out of 4.00

Given the following list, provide where within a computer they would be located.

Stack Pointer	CPU	✓
Program Counter	CPU	✓
Processor Exception handler "main method"	Kernel Stack	✗
Process Control Block	Kernel Stack	✓
UpCall handler	User Instructions	✓
Operating System's 'main method'	User Instructions	✗
Interrupt Stack	Kernel Stack	✓
IVT Address Register	CPU	✓

The correct answer is:

Stack Pointer → CPU,

Program Counter → CPU,

Processor Exception handler "main method" → Kernel Instructions,

Process Control Block → Kernel Stack,

UpCall handler → User Instructions,

Operating System's 'main method' → Kernel Instructions,

Interrupt Stack → Kernel Stack,

IVT Address Register → CPU

Question **12**

Correct

Mark 1.00 out of 1.00

Given the code below answer the following questions

```
int a = 84;
int fd = open(...) //opening a file
int ret = fork();
if(ret > 0) {
    close(fd);
    a = 34;
    ...
}else if(ret == 0) {
    printf("a=%d\n", a);
    read(fd, something);
}
```

After the new process is forked, suppose that the parent process is scheduled first, before the child after fork, it closes the file descriptor and changes the value of a variable as shown above. Assume for the first time only after the parent completes these two changes

What is the value of the variable ``a" as printed in the child process, when it is scheduled next?

Answer: 

The correct answer is: 84.00

Question **13**

Correct

Mark 4.00 out of 4.00

You are told that a given system uses Segmentation. The [memory](#) is currently full and holds 3 processes: process A with a size of 512KB as well as the kernel whose size is 1024KB. You can assume there is nothing else that is taking up space. You are told that the base of the stack which grows upwards for process B is found at the very end of the Address Space. You are also told that the last portion of the Address Space for each process is used for the Stack.

After the Context switch which process does the values in the segment register correspond to?

What is the Physical Address of the base of the stack for Process B? (Just give the value in KB)

What is the offset of the Virtual Address 500KB in KB? (Just give the value) 

What is the Physical Address that the Virtual Address 500KB maps to in KB? (Just give the value)

Question **14**

Correct

Mark 2.00 out of 2.00

A system uses a simple base/bounds to virtualize address spaces.

Determine the values of the base and bounds which would result in the following address translation.

You can answer using either of the bounds implementations as discussed in class.

Translations

Virtual Address	Physical Address
0	500
100	600
2499	2999
2500	[fault]

Base = ✓Bounds = ✓

Question **15**

Correct

Mark 1.00 out of 1.00

If the parent process has a PID of 1000 and the fork() function is called, which of the following is true?

Select one:

- ☒ a. The child process will have a PID greater than 1000. ✓
- ☐ b. The child process cannot run until the parent process finishes.
- ☐ c. The child process takes over the role of the parent process.
- ☐ d. The child process will always have a PID of 1001.
- ☐ e. The child process will inherit the PID of the parent process.

The correct answer is:

The child process will have a PID greater than 1000.

[◀ Test 1 Part Time](#)

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